

Hankel Determinants of Geometric Moments and Rational Lambert Values

Erdős Problem #1049

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ABSTRACT

We determine the fixed-base asymptotic of Zudilin's normalised Hankel determinants, including the constant and the factor $N^{-8F(1/q)}$, where $F(t) = \sum_{n \geq 1} (t^n - 1)^{-1}$. The proof uses positive geometric moments and a rank-uniform partition estimate. Separately, we extend the rational-base irrationality region for F by calculating the denominator cost of Zudilin's 2004 forms. The region includes every positive integral power of $31/4$, but excludes $3/2$.

1 Introduction

For each fixed real $q \in (0, 1)$, we determine the asymptotic size of the Hankel determinants arising from Zudilin's q -logarithm approximations. Write $(z; q)_m = \prod_{j=0}^{m-1} (1 - zq^j)$ and $F(t) = \sum_{n \geq 1} (t^n - 1)^{-1}$ for $t > 1$. At the auxiliary parameters $x = z = 1$, the normalisation in [5, (6), §4] gives

$$v_m^*(q) = \sum_{t \geq 0} q^{(m+1)t} \frac{(q; q)_m^3 (q^{t+1}; q)_m}{(q^{m+1+t}; q)_{m+1}}, \quad V_N^*(q) = \det(v_{i+j}^*(q))_{0 \leq i, j < N}. \quad (1)$$

The sums converge for $0 < q < 1$. They also define formal series, since the t th summand has q -order $(m+1)t$.

Theorem 1.1 (the size of V_N^* at a fixed base). *Fix $0 < q < 1$ and write $P = (q; q)_\infty$, $B_N = N(N-1)(2N-1)/6$ and $C_N = (N!)^2(N+1)!/2^N$. There is a $K(q) > 0$ with*

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$$V_N^*(q) \sim K(q) C_N q^{B_N} P^{2N} N^{-8F(1/q)} \quad (N \rightarrow \infty).$$

The constant $K(q)$ is the convergent product in (9). Zudilin proved $\text{ord}_q V_N^* \geq B_N$ and the estimate $|V_N^*| \leq q^{N^3/3} \exp(O_q(N^2))$ [5, Lemma 1 and §4]. We obtain the factorial and exponential factors concealed by this estimate, as well as its power of N . We also recover the exact first formal term $C_N q^{B_N}$. These are different limiting questions:

**Authorship and AI use.* Will Cook built and directed the research infrastructure and reviewed the claims when he could. AI agents did most of the research and drafting. Cook did not independently verify every claim.

multiplying by $(1 - q)^{N^3}$ preserves that first term and changes the fixed-base logarithm by a cubic quantity. No uniformity as $q \rightarrow 1$ is asserted.

We first express v_m^* as a positive moment on $1, q, q^2, \dots$. Heine's identity then expands V_N^* into squared Vandermonde products, as in [7, §2, (2)–(5)]. After division by the contribution from the first N nodes, the remaining tuples are indexed by partitions. A summable bound uniform in N reduces their limit to a Cauchy determinant. The weight calculation supplies the remaining factor: $a_k/c_k = 1 - 8F(1/q)/(k+1) + O_q((k+1)^{-2})$. Its product over $k < N$ accounts for $N^{-8F(1/q)}$. The geometric-moment argument is stated independently in Theorem 2.1; for example, it also evaluates the determinant with entries $(1 - q^{i+j+1})^{-2}$.

Chowla's conjecture, recorded by Erdős [1, p. 102], asks whether $F(t)$ is irrational for every rational $t > 1$. This is a related arithmetic question, and the fixed-base asymptotic alone does not answer it. In Section 4 we use the different 2004 construction [4, §5] to prove irrationality for coprime $a > b \geq 1$ satisfying

$$\frac{\log b}{\log a} < \theta^* = 0.4056830213840605 \dots,$$

where (10) defines θ^* exactly. The calculation cancels the common polynomial factors before clearing denominators at a/b . It includes every positive integral power of $31/4$. Neither this region nor the determinant argument settles the case $3/2$.

Section 2 proves the general determinant formula. Section 3 constructs the weights and proves Theorem 1.1. The rational-base argument is self-contained in Section 4, with longer calculations and complementary questions linked in Section 5. All logarithms are natural, empty products and determinants equal 1, and constants in $O_q(\cdot)$ may depend on the fixed q .

2 Geometric moment determinants

The determinant argument depends on the relative growth of the weights. The following hypotheses allow polynomial growth, including decaying powers, and suffice for the weights in (1). Put

$$P = (q; q)_\infty, \quad M(q) = \prod_{d \geq 1} (1 - q^d)^{-d}, \quad B_N = \frac{N(N-1)(2N-1)}{6}.$$

The infinite products are positive and finite because their logarithms have summable tails.

Theorem 2.1 (geometric moment determinants). *Let $0 < q < 1$ and $a_k > 0$ for $k \geq 0$. Suppose that, for fixed constants $C > 0$ and $\kappa \geq 0$,*

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$$\frac{a_{k+h}}{a_k} \rightarrow 1 \quad (k \rightarrow \infty) \text{ for each fixed } h \geq 0, \quad \frac{a_{k+h}}{a_k} \leq C(1+h)^\kappa \quad (k, h \geq 0). \quad (2)$$

For $M_m = \sum_{k \geq 0} a_k q^{(m+1)k}$ and $D_N = \det(M_{i+j})_{0 \leq i, j < N}$, we have

$$D_N \sim M(q)^3 q^{B_N} P^{2N} \prod_{k=0}^{N-1} a_k \quad (N \rightarrow \infty). \quad (3)$$

For every real s , the weights $a_k = (k+1)^s$ satisfy (2) with $C = 1$ and $\kappa = \max(s, 0)$, and their first N terms have product $(N!)^s$. Taking $s = 1$ and summing the moments gives

$$\det \left(\frac{1}{(1 - q^{i+j+1})^2} \right)_{0 \leq i, j < N} \sim M(q)^3 N! q^{B_N} P^{2N}. \quad (4)$$

Proof of Theorem 2.1. The bound $a_h \leq Ca_0(1+h)^\kappa$ ensures convergence of every moment. Apply Cauchy–Binet to a finite set of nodes and let its size tend to infinity. The matrix entries converge, while positivity permits passage to the limit in the sum, giving

$$D_N = \sum_{0 \leq k_0 < \dots < k_{N-1}} \prod_{i < N} a_{k_i} q^{k_i} \prod_{i < j < N} (q^{k_i} - q^{k_j})^2. \quad (5)$$

The tuple $k_i = i$ contributes

$$q^{B_N} \Delta_N \prod_{i < N} a_i, \quad \Delta_N = \prod_{d=1}^{N-1} (1 - q^d)^{2(N-d)}.$$

For any other tuple, write $k_i = i + \lambda_i$ and put $\mu_j = \lambda_{N-j}$ for $1 \leq j \leq N$. Then $\mu_1 \geq \dots \geq \mu_N \geq 0$. Division by the contribution of $k_i = i$ turns its summand into

$$W_N(\mu) = q^{\sum_{j \leq N} (2j-1)\mu_j} \prod_{j \leq N} \frac{a_{N-j+\mu_j}}{a_{N-j}} \cdot \prod_{1 \leq j < k \leq N} \left(\frac{1 - q^{k-j+\mu_j-\mu_k}}{1 - q^{k-j}} \right)^2. \quad (6)$$

Set $W_N(\mu) = 0$ when μ has more than N positive parts, so that the index set of the sum is independent of N .

For a partition of length ℓ , only the factors with $j \leq \ell$ can differ from 1. For each such j , the denominator product in the Vandermonde quotient is at least P , and its numerator product is at most 1. Consequently

$$W_N(\mu) \leq (CP^{-2})^\ell \prod_{j=1}^{\ell} (1 + \mu_j)^\kappa q^{(2j-1)\mu_j}.$$

Let $A = \sum_{u \geq 1} (1+u)^\kappa q^{u-1} < \infty$. Dropping the ordering of the positive parts, we obtain

$$\begin{aligned} \sum_{\ell(\mu)=\ell} \sup_N W_N(\mu) &\leq (CP^{-2})^\ell \prod_{j=1}^{\ell} \sum_{u \geq 1} (1+u)^\kappa q^{(2j-1)u} \\ &\leq (CP^{-2}A)^\ell q^{\ell^2}. \end{aligned}$$

The last inequality follows from $q^{(2j-1)u} \leq q^{2j-1} q^{u-1}$. Summing over ℓ gives a finite bound, uniform in the determinant rank.

For each fixed partition, all the weight ratios in (6) tend to 1. Its Vandermonde quotient also converges: there are only finitely many nontrivial first indices, and the logarithms of the remaining factors have geometric tails. Dominated convergence therefore

reduces the limit of $\sum_{\mu} W_N(\mu)$ to the case $a_k = 1$. For these weights $M_m = (1 - q^{m+1})^{-1}$, and Cauchy's determinant formula gives

$$D_N = \frac{q^{B_N} \Delta_N}{\prod_{0 \leq i, j < N} (1 - q^{i+j+1})}.$$

The multiplicity of $1 - q^d$ in the denominator increases to d , so the normalised sum tends to $\mathcal{M}(q)$. Finally,

$$\frac{\Delta_N}{p^{2N}} = \prod_{d < N} (1 - q^d)^{-2d} \prod_{d \geq N} (1 - q^d)^{-2N} \rightarrow \mathcal{M}(q)^2,$$

since the logarithm of the second product is $O_q(Nq^N)$. These two limits prove (3). $\square$

3 The weights of Zudilin's moments

We apply Theorem 2.1 by collecting the dependence on m in (1) into the single variable $w = q^{m+1}$. For this purpose, set

$$G_q(w) = \frac{1}{(w; q)_{\infty}^3} \sum_{t \geq 0} \frac{w^t}{(q; q)_t} \frac{(q^t w^2; q)_{\infty}}{(q^t w; q)_{\infty}^2}, \quad a_k(q) = [w^k] P^4 G_q(w). \quad (7)$$

The product identities $(q; q)_m = P / (q^{m+1}; q)_{\infty}$ and $(q^a; q)_m = (q^a; q)_{\infty} / (q^{a+m}; q)_{\infty}$ give

$$v_m^* = P^4 G_q(q^{m+1}) = \sum_{k \geq 0} a_k(q) q^{(m+1)k}. \quad (8)$$

These identities hold formally in $\mathbb{Z}[[q, w]]$ and, for fixed $0 < q < 1$, analytically for $|w| < 1$. In the formal identity only finitely many t contribute to any w -coefficient. For the analytic identity, we use the absence of denominator zeros in that disc and the locally uniform bounds on the product tails. The factor w^t then gives locally uniform convergence of the sum. To prove positivity and obtain a bound uniform in the shift, we express its coefficients as finite sums.

3.1 Positive weights

For $r \geq 1$ and $k \geq 0$ define the finite sum

$$R_k^{(r)}(q) = \sum_{n_1 + \dots + n_r = k} \frac{(q; q)_k}{\prod_{j=1}^r (q; q)_{n_j}},$$

over r nonnegative parts. These are the multivariate Rogers–Szegő polynomials at unit arguments, written as sums of Gaussian multinomial polynomials [19]. At $q = 0$ they count compositions, and

$$c_k := R_k^{(2)}(0) R_k^{(3)}(0) = \frac{(k+1)^2 (k+2)}{2}, \quad \prod_{k < N} c_k = C_N.$$

Proposition 3.1 (the positive moment weights). *For every $k \geq 0$,*

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$$a_k = P^4 \frac{R_k^{(2)} R_k^{(3)}}{(q; q)_k}.$$

At fixed $0 < q < 1$ these weights satisfy

$$P^5 c_k \leq a_k \leq P^{-1} c_k, \quad \frac{a_{k+h}}{a_k} \leq P^{-6} (1+h)^3 \quad (k, h \geq 0).$$

Proof. We use the Gaussian binomial coefficients $\begin{bmatrix} n \\ j \end{bmatrix}_q = (q; q)_n / ((q; q)_j (q; q)_{n-j})$ for $0 \leq j \leq n$, and put $\mathcal{E}(z) = (z; q)_\infty^{-1}$. For $\partial_q f(z) = (f(z) - f(qz))/z$ we have $\partial_q \mathcal{E} = \mathcal{E}$ and $\mathcal{E}(z)^r = \sum_{n \geq 0} R_n^{(r)} z^n / (q; q)_n$ by Euler's expansion. The q -Leibniz rule, which follows by induction from the product rule, is

$$\partial_q^n (fg)(z) = \sum_{j=0}^n \begin{bmatrix} n \\ j \end{bmatrix}_q (\partial_q^j f)(q^{n-j} z) (\partial_q^{n-j} g)(z).$$

Let $h_k(z) = \sum_{j=0}^k \begin{bmatrix} k \\ j \end{bmatrix}_q (z; q)_j$. Applying the rule to $\mathcal{E} \cdot \mathcal{E}$ and using $\mathcal{E}(q^m z) = (z; q)_m \mathcal{E}(z)$, we obtain $\partial_q^k \mathcal{E}(z)^2 = \mathcal{E}(z)^2 h_k(z)$ (for $k = 1$, both sides equal $\mathcal{E}(z)^2 (2 - z)$). A second application, this time to $\mathcal{E} \cdot \mathcal{E}^2$, gives

$$\partial_q^n \mathcal{E}(z)^3 = \mathcal{E}(z)^3 \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix}_q (z; q)_k h_k(z),$$

and hence

$$\sum_{n \geq 0} \frac{w^n}{(q; q)_n} \partial_q^n \mathcal{E}(z)^3 = \mathcal{E}(w) \mathcal{E}(z)^3 \sum_{k \geq 0} \frac{w^k (z; q)_k}{(q; q)_k} h_k(z).$$

We now put $z = w$. The coefficient of w^n on the left is $R_n^{(3)} \sum_{j \leq n} ((q; q)_j (q; q)_{n-j})^{-1} = R_n^{(2)} R_n^{(3)} / (q; q)_n$. To evaluate the right-hand side, we expand h_k , put $k = j + t$, and use the q -binomial theorem in the form

$$\sum_{t \geq 0} \frac{(wq^j; q)_t}{(q; q)_t} w^t = \frac{(w^2 q^j; q)_\infty}{(w; q)_\infty}.$$

The resulting expression is $G_q(w)$, as required. We may read each identity coefficient-wise, or as an absolutely convergent expansion when $|z|$ and $|w|$ are sufficiently small.

For the bounds, we write $b_k^{(r)} = [z^k](z; q)_\infty^{-r}$, so that $R_k^{(r)} = (q; q)_k b_k^{(r)}$ and $a_k = P^4 (q; q)_k b_k^{(2)} b_k^{(3)}$. By Euler's expansion, $b_k^{(r)}$ is an r -fold convolution of $1/(q; q)_n$, whose terms lie in $[1, P^{-1}]$. There are $\binom{k+r-1}{r-1}$ compositions of k into r nonnegative parts, and we deduce

$$\binom{k+r-1}{r-1} \leq b_k^{(r)} \leq P^{-r} \binom{k+r-1}{r-1}.$$

Combining these at $r = 2, 3$ with $P \leq (q; q)_k \leq 1$ and $c_k = (k+1) \binom{k+2}{2}$ gives $P^5 c_k \leq a_k \leq P^{-1} c_k$. The shift bound follows from $c_{k+h}/c_k \leq (1+h)^3$. $\square$

The bounds above control shifts uniformly. We now calculate the first correction to a_k/c_k ; this will give the fixed-shift limit and the power of N in Theorem 1.1.

3.2 The coefficient asymptotic

Proof of Theorem 1.1. Put $L = F(1/q)$, $e_k = (q; q)_k^{-1}$ and $E = P^{-1}$. We need the first correction to the limiting value $e_k \rightarrow E$. Writing $d_k = E - e_k$, the identity $e_{k+1} - e_k = q^{k+1}e_{k+1}$ gives

$$d_k = \sum_{j>k} q^j e_j, \quad 0 \leq d_k \leq \frac{Eq^{k+1}}{1-q}.$$

Thus $\sum d_k$ and $\sum kd_k$ converge. To evaluate the former, use Euler's identity $\mathcal{E}(z) = \sum_{k \geq 0} e_k z^k = (z; q)_\infty^{-1}$ for $|z| < 1$. Tonelli's theorem and logarithmic differentiation at $z = q$ give

$$S := \sum_{k \geq 0} d_k = \sum_{j \geq 1} j q^j e_j = q \mathcal{E}'(q) = E \sum_{r \geq 1} \frac{q^r}{1 - q^r} = EL.$$

The differentiation is justified by locally uniform convergence on $|z| < 1$.

Let $b_k^{(r)} = [z^k] \mathcal{E}(z)^r$. The finite-sum formula in Proposition 3.1 becomes

$$a_k = P^4 (q; q)_k b_k^{(2)} b_k^{(3)}.$$

We expand each finite convolution using $e_k = E - d_k$. For two factors,

$$b_k^{(2)} = E^2(k+1) - 2E \sum_{i=0}^k d_i + \sum_{i=0}^k d_i d_{k-i} = E^2(k+1 - 2L) + O_q((k+1)q^k).$$

For three factors, the terms with one d_i contribute $-3E^2 \sum_{i=0}^k (k-i+1)d_i$. Here the sum equals $(k+1)S + O_q(1)$ because $\sum id_i < \infty$. Terms with two d_i are bounded by $3ES^2$, and the term with three is $O_q((k+1)^2 q^k)$. It follows that

$$b_k^{(3)} = E^3 \left(\binom{k+2}{2} - 3L(k+1) \right) + O_q(1).$$

Since $(q; q)_k = P(1 + O_q(q^k))$ and $c_k = (k+1)^2(k+2)/2$, multiplication yields

$$\begin{aligned} \frac{a_k}{c_k} &= \left(1 - \frac{2L}{k+1} + O_q(q^k) \right) \left(1 - \frac{6L}{k+2} + O_q((k+1)^{-2}) \right) (1 + O_q(q^k)) \\ &= 1 - \frac{8L}{k+1} + O_q((k+1)^{-2}). \end{aligned}$$

In particular $a_{k+1}/a_k \rightarrow 1$, and multiplication of finitely many adjacent ratios gives $a_{k+h}/a_k \rightarrow 1$ for each fixed h . The uniform shift bound follows from Proposition 3.1, so Theorem 2.1 applies.

Let γ_E denote Euler's constant and put

$$A(q) = e^{-8\gamma_E L} \prod_{k \geq 0} \left(\frac{a_k}{c_k} e^{8L/(k+1)} \right), \quad K(q) = A(q)M(q)^3. \quad (9)$$

Positivity of a_k and summability of $\log(a_k/c_k) + 8L/(k+1) = O_q((k+1)^{-2})$ show that $A(q)$ converges to a positive number. Using $\sum_{k < N} (k+1)^{-1} = \log N + \gamma_E + o(1)$, we obtain

$$\prod_{k < N} a_k \sim A(q) C_N N^{-8L}.$$

Substitution in (3) proves the theorem, or equivalently

$$\log V_N^* = B_N \log q + \log C_N + 2N \log P - 8L \log N + \log K(q) + o(1).$$

□

The power of N thus comes from the first-order corrections in the two convolutions. Replacing a_k by its leading cubic term c_k would lose both corrections and hence the factor $N^{-8F(1/q)}$.

3.3 The first nonzero formal term

The order of a nonzero formal series is the least exponent with nonzero coefficient. In Heine's expansion for V_N^* , a single tuple attains the least order.

Theorem 3.2 (the first nonzero term of the Hankel determinant). *For every $N \geq 1$,*

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$$\text{ord}_q V_N^* = \frac{N(N-1)(2N-1)}{6},$$

and the coefficient of the first nonzero monomial is

$$[q^{N(N-1)(2N-1)/6}] V_N^* = \frac{(N!)^2 (N+1)!}{2^N}.$$

Proof. At $q = 0$, the $t = 0$ term of $G_q(w)$ is $(1 - w^2)/(1 - w)^5$, while the remaining terms sum to $w/(1 - w)^4$. Thus

$$G_0(w) = \frac{1 + 2w}{(1 - w)^4}, \quad a_k(0) = c_k > 0.$$

We compute modulo q^{M+1} by truncating each moment sum in (8) at $k = M$. Since all omitted terms are divisible by q^{M+1} , multilinearity leaves the determinant unchanged modulo that power. Applying finite Cauchy–Binet and then allowing M to increase, we obtain the coefficientwise identity

$$V_N^* = \sum_{k_0 < \dots < k_{N-1}} \left(\prod_{i < N} a_{k_i}(q) q^{k_i} \right) \prod_{i < j < N} (q^{k_i} - q^{k_j})^2.$$

The order of the summand indexed by (k_i) is $\sum_{i < N} (2N - 1 - 2i)k_i$. Because $k_i \geq i$ and all the weights $2N - 1 - 2i$ are positive, we obtain the unique minimum at $k_i = i$, with order and coefficient

$$\sum_{i < N} (2N - 1 - 2i)i = B_N, \quad \prod_{i < N} c_i = C_N.$$

No other tuple can cancel that coefficient. For example, at rank two the pair $(0, 1)$ contributes $6q + O(q^2)$, and every other increasing pair has order at least two. □

This proves equality in Zudilin's bound [5, §4, pp. 6–7]. An alternative argument, which computes the first nonzero term of every transformed row, is given in [the companion paper](#).

4 Rational bases

We now use the 2004 coefficient family. If $U, V \in \mathbb{Z}[X]$ have degree at most W , evaluation at a/b requires the factor b^W to produce integer coefficients. A common polynomial factor should therefore be cancelled first. For example, at $3/2$ the pair $(X+1)(X+2), (X+1)(X+3)$ gives $(35, 45)$ after clearing to degree 2. After cancelling $X+1$, clearing to degree 1 gives $(7, 9)$. For Zudilin's forms, cyclotomic cancellation has a degree of quadratic order, and this degree determines the rational-base criterion.

We use the parameter direction from Zudilin's construction [4, §5, pp. 161–162], together with its thirteen intervals and constants C_1, C_0 . At integer bases his theorem gives the irrationality-exponent bound $C_1/C_0 = 2.46497868\dots$ [4, Thm. 1, p. 154]. Let $\psi_1(u) = \sum_{k \geq 0} (k+u)^{-2}$ for $u > 0$, and let $\mathcal{I}$ consist of the thirteen intervals listed in the proof. We put

$$C_1 = \frac{1091}{2}, \quad J = \sum_{[u,v] \in \mathcal{I}} (\psi_1(u) - \psi_1(v)), \quad C_0 = 266 - \frac{3}{\pi^2}(225 - J). \quad (10)$$

The intervals are disjoint and lie in $[1/14, 1)$, so $0 \leq J \leq \psi_1(1/14) - \psi_1(1) < 196$. Together with $\pi > 3$, these bounds give $0 < C_0 < 266 < C_1/2$. Thus $\theta^* = C_0/C_1$ and $\mu = C_1/C_0$ are positive reciprocal constants. The notation $\mu_{\text{irr}}(\xi)$ instead denotes the irrationality exponent of a value. The estimates below give a sufficient cutoff; its optimality is unknown.

Theorem 4.1 (rational-base region for Zudilin's forms). *Let $a > b \geq 1$ be coprime integers with*

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$$b^\mu < a, \quad \text{equivalently} \quad \frac{\log b}{\log a} < \theta^*,$$

$$\theta^* = \frac{C_0}{C_1} = 0.4056830213840605\dots,$$

$$\mu = \frac{C_1}{C_0} = 2.4649786835749750\dots$$

Then $F(a/b)$ is irrational.

The condition holds for every integer base, since $b = 1$ gives $\log b / \log a = 0$. For a fixed denominator $b > 1$, it requires $a > b^{2.46497868\dots}$, much more than $a > b$; nevertheless it gives infinitely many reduced noninteger rational bases for each such b . The base $31/4$ satisfies the condition, and taking a common positive integer power of the numerator and denominator leaves their logarithmic ratio unchanged. Since $\theta^* < 1/2 < \log 2 / \log 3$, the condition fails at $3/2$. At equality $\log b / \log a = \theta^*$ the quadratic exponent vanishes, so the estimates below give no conclusion.

Proof of Theorem 4.1. We shall calculate the degree W_n of the cancelled coefficient and show that $b^{W_n} \Lambda_n(a/b)$ is positive and tends to zero. We fix $n \geq 1$ and set

$$a_0 = 14n + 1, \quad a_1 = 12n + 1, \quad a_2 = 14n + 1, \quad \beta = 27n + 2, \quad N = 15n.$$

In this proof N denotes the cyclotomic cutoff. We use the coefficient pair A_n, B_n of the source identities [4, (8)–(11), pp. 156–157], with $H_n = A_n F - B_n$. Let Φ_ℓ be the cyclotomic polynomial whose roots are the primitive ℓ th roots of unity. Put $D_N(X) = \prod_{\ell=1}^N \Phi_\ell(X)$, $M_n = 266n^2 + 34n + 1$, and

$$\Omega_n(X) = \prod_{\ell=2}^N \Phi_\ell(X)^{\nu_\ell}, \quad \nu_\ell = \omega(n/\ell),$$

where the periodic function

$$\omega(x) = \max\{0, \lfloor 14x \rfloor + \lfloor 13x \rfloor - \lfloor 12x \rfloor - \lfloor 15x \rfloor, \\ 2\lfloor 14x \rfloor - \lfloor 13x \rfloor - \lfloor 15x \rfloor\}$$

has period 1: the coefficients of the floor arguments in each difference have equal sums, $14 + 13 = 12 + 15$ and $2 \cdot 14 = 13 + 15$. On $[0, 1)$ it is zero or one, with support

$$\begin{aligned} &[1/14, 1/12), [1/7, 1/6), [3/14, 1/4), [2/7, 1/3), \\ &[5/14, 2/5), [3/7, 7/15), [1/2, 8/15), [4/7, 3/5), \\ &[9/14, 2/3), [5/7, 11/15), [11/14, 4/5), \\ &[6/7, 13/15), [13/14, 14/15). \end{aligned}$$

These are the intervals $\mathcal{I}$ used to define J . The function ω , these thirteen intervals and the exponents $\nu_\ell = \omega(n/\ell)$ of the source's (22) are printed at [4, pp. 161–162].

We first bound the remainder before cancellation. For $x > 1$ we put $q = 1/x$ and use identity (9) on p. 156 of the source, which gives the positive representation

$$H_n(x) = \sum_{t \geq 0} q^{a_0 t} \frac{(q^{t+1}; q)_{a_1-1}}{(q; q)_{a_1-1}} \frac{(q; q)_{\beta-a_2-1}}{(q^{a_2+t}; q)_{\beta-a_2}}.$$

The denominator length is $\beta - a_2$, as in the gamma expression (7) and residues (8) of [4, p. 156]; the unnumbered display of $R(T)$ on that page has length $\beta - a_2 - 1$ instead. Each of the four finite products displayed above lies between $P = (q; q)_\infty > 0$ and 1, so

$$P^2 \leq H_n(x) \leq \frac{P^{-2}}{1 - q^{a_0}}.$$

In particular $H_n(x) > 0$ and $\log H_n(x) = O_x(1)$.

To identify the integral polynomials we use the inclusion in Zudilin's Lemma 7, display (23) [4, p. 161]. The proof compares the cyclotomic valuations of the coefficient polynomials under six admissible parameter permutations, and gives the divisibility in $\mathbb{Z}[X]$ that we need before evaluation. Its parameter vector is $n(13, 14, 12, 14, 15, 13)$, its maximum is $15n$, and $\beta - a_1 - a_2 = n > 0$, which is the positivity condition $s > 0$ of the lemma. The conditions (14) of [4, p. 157], $a_1 \leq a_2$ and $a_1 + a_2 \leq \beta \leq a_0 + a_2$, hold as well, so the exponent in (23) is the integer (16), which equals M_n . Lemma 7 therefore gives, for every $n \geq 1$,

$$\Lambda_n(X) = X^{-M_n} \frac{D_N(X)}{\Omega_n(X)} H_n(X) = U_n(X) F(X) - V_n(X), \quad U_n, V_n \in \mathbb{Z}[X].$$

This is the polynomial conclusion before integer specialisation in the source's (24), p. 162. The coefficients are the cancelled source coefficients:

$$U_n = X^{-M_n}(D_N/\Omega_n)A_n, \quad V_n = X^{-M_n}(D_N/\Omega_n)B_n.$$

Indeed, coefficients of F and 1 over $\mathbb{Q}(X)$ are unique because F is not a rational function. To see this, we split $F(e^h)$ at $m = 1/h$ as $h \downarrow 0$. For $m \leq 1/h$, the inequalities $y^{-1} - 1 \leq (e^y - 1)^{-1} \leq y^{-1}$ for $0 < y \leq 1$ give $h^{-1} \sum_{m \leq 1/h} m^{-1} + O(h^{-1})$. Bounding the remaining terms by a geometric tail gives a further $O(h^{-1})$, and therefore $F(e^h) = h^{-1} \log(1/h) + O(h^{-1})$, which has no rational-function pole order at $X = 1$. The cancelled remainder is positive for $x > 1$, since D_N/Ω_n is a product of cyclotomic polynomials positive there.

The Gaussian binomial polynomial $\begin{bmatrix} m \\ k \end{bmatrix}_X = \prod_{j=1}^k (1 - X^{m-k+j})/(1 - X^j)$ is monic of degree $k(m - k)$ for integers $0 \leq k \leq m$. The source's (8) and (10) on p. 156 give the coefficient whose degree we need:

$$A_n(X) = \sum_{k=a_2}^{\beta-1} (-1)^{a_1+a_2+k+1} X^{a_0k + \binom{a_1}{2} - \binom{\beta-a_2}{2} + \binom{\beta-k}{2}} \begin{bmatrix} k-1 \\ a_1-1 \end{bmatrix}_X \begin{bmatrix} \beta-a_2-1 \\ \beta-k-1 \end{bmatrix}_X.$$

On comparing successive summands, we find that the degree increases by $40n+1-k > 0$ for $a_2 \leq k \leq \beta - 2$. The last summand alone therefore determines the leading term, giving

$$K_n := \deg A_n = \frac{1091n^2 + 81n + 2}{2}, \quad W_n := \deg U_n = K_n - M_n + \sum_{\ell \leq 15n} (1 - \nu_\ell) \varphi(\ell).$$

Here $\varphi(\ell) = \deg \Phi_\ell$ is Euler's totient function, and $\nu_1 = \omega(n) = 0$. The bounds for H_n are uniform for $x \geq 2$: then $P \geq \prod_{j \geq 1} (1 - 2^{-j}) > 0$ and $(1 - x^{-a_0})^{-1} \leq 2$. For fixed n , therefore, $H_n(x) = O(1)$ and $\Lambda_n(x) = O(x^{W_n - K_n})$ as $x \rightarrow \infty$. The unique top summand of A_n has leading coefficient $(-1)^{a_1+a_2+\beta} = (-1)^n$. Since the Gaussian factors and D_N/Ω_n are monic, U_n has that leading coefficient too. Moreover $F(x) = x^{-1} + O(x^{-2})$ and $K_n \geq 2$, so

$$V_n(x) = U_n(x)F(x) - \Lambda_n(x) = (-1)^n x^{W_n-1} + O(x^{W_n-2}).$$

Here $W_n \geq K_n - M_n = (559n^2 + 13n)/2 > 0$. Thus $\deg V_n = W_n - 1$, with the same unit leading coefficient as U_n . For every prime $p \mid b$, reduction of the cleared first coordinate gives

$$b^{W_n} U_n(a/b) \equiv (-1)^n a^{W_n} \not\equiv 0 \pmod{p}.$$

Thus $U_n(a/b)$ has reduced denominator exactly b^{W_n} , the least common clearing denominator for this pair. Every common integer divisor of the cleared row is consequently coprime to b . This leaves common factors at other primes and savings from combinations of different rows to be analysed separately.

Having computed the degree by keeping n fixed and letting $x \rightarrow \infty$, we now estimate decay with x fixed and $n \rightarrow \infty$.

It remains to find the quadratic part of the degree removed by cancellation. The limits below are the cyclotomic limits of [4, Lemmas 1–2, p. 155]. The proof uses the argument through reciprocal intervals, the summatory totient estimate and the trigamma

function from the proof of [3, Lemma 1, p. 466], with an explicit truncation of the block sum. The elementary summatory estimate $\sum_{\ell \leq y} \varphi(\ell) = 3y^2/\pi^2 + O(y \log y)$ gives

$$\frac{1}{n^2} \sum_{\ell \leq 15n} \varphi(\ell) \rightarrow \frac{675}{\pi^2}.$$

For $[u, v] \in \mathcal{I}$, the condition $\{n/\ell\} \in [u, v)$ is the disjoint union of intervals $n/(k+v) < \ell \leq n/(k+u)$ for $k \geq 0$. We may apply the same summatory estimate term by term to any finite set of k . If we stop before $k = K$, with $K \geq 1$, every omitted block has $\ell \leq n/(K+u)$. Thus the elementary bound $\sum_{\ell \leq y} \varphi(\ell) \leq y^2$ for $y \geq 0$ gives the uniform estimate

$$\frac{1}{n^2} \sum_{\ell \leq n/(K+u)} \varphi(\ell) \leq \frac{1}{(K+u)^2}.$$

Indeed, for $y \geq 1$ one can bound $\varphi(\ell)$ by ℓ and sum; for $0 \leq y < 1$ the sum is empty. This bound is uniform in n . We can now let n and then K tend to infinity, obtaining

$$\frac{1}{n^2} \sum_{\ell \leq 15n} v_\ell \varphi(\ell) \rightarrow \frac{3}{\pi^2} \sum_{[u,v] \in \mathcal{I}} \sum_{k \geq 0} \left(\frac{1}{(k+u)^2} - \frac{1}{(k+v)^2} \right) = \frac{3J}{\pi^2}.$$

Here no contributing index exceeds $14n$, since $u \geq 1/14$. Consequently

$$K_n/n^2 \rightarrow C_1, \quad (K_n - W_n)/n^2 \rightarrow C_0.$$

We return to a fixed real base $x > 1$ and estimate the cyclotomic product. Writing μ_{Mob} for the Möbius function, we have

$$\log \Phi_\ell(x) - \varphi(\ell) \log x = \sum_{d|\ell} \mu_{\text{Mob}}(d) \log(1 - x^{-\ell/d}).$$

The total absolute error over $\ell \leq 15n$ is $O_x(n)$, since it is at most

$$15n \sum_{d \geq 1} \frac{-\log(1 - x^{-d})}{d}.$$

The series converges because $-\log(1 - x^{-d}) \leq x^{-d}/(1 - x^{-1})$. Since $0 \leq 1 - v_\ell \leq 1$, it follows that

$$\log \Lambda_n(x) = -(K_n - W_n) \log x + O_x(n).$$

We can now clear the denominators at the rational base. Since $U_n, V_n \in \mathbb{Z}[X]$ and their degrees are at most W_n , the numbers $b^{W_n} U_n(a/b)$ and $b^{W_n} V_n(a/b)$ are integers. Moreover

$$\begin{aligned} \log(b^{W_n} \Lambda_n(a/b)) &= K_n \log b - (K_n - W_n) \log a + O_{a/b}(n) \\ &= (C_1 \log b - C_0 \log a) n^2 + o(n^2). \end{aligned}$$

The coefficient is negative under the theorem's hypothesis. The positive values of these integer-coefficient forms therefore tend to zero. If $F(a/b) = r/s$ with integers r, s and $s \neq 0$, every nonzero value would have absolute value at least $1/|s|$, a contradiction. $\square$

Corollary 4.2. $F((31/4)^r)$ is irrational for every integer $r \geq 1$.

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Proof. The exact inequalities $31^2 < 4^5$ and $4^{200} < 31^{81}$ give $2/5 < \log 4 / \log 31 < 81/200$. The first term of each trigamma difference yields

$$J \geq \sum_{[u,v] \in \mathcal{I}} (u^{-2} - v^{-2}) = \frac{2015640690251}{25971865920}.$$

Using $\pi > 157/50$ gives the rational lower bound

$$\theta^* > \frac{2359630009523263}{5820307922172744} > \frac{81}{200}.$$

Taking a common positive integer power preserves both the logarithmic ratio and coprimality, so the same criterion applies for every $r \geq 1$. $\square$

Corollary 4.3 (an irrationality measure uniform over powers). *For coprime $a > b \geq 1$ with $\theta = \log b / \log a < \theta^*$ and every integer $r \geq 1$,*

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$$\mu_{\text{irr}}(F((a/b)^r)) \leq \frac{1 - \theta}{\theta^* - \theta}.$$

Here $\mu_{\text{irr}}(\xi)$ is the supremum of the exponents ν for which $|\xi - p/q| < q^{-\nu}$ has infinitely many reduced rational solutions. In particular, $\mu_{\text{irr}}(F((31/4)^r)) < 301$ for every $r \geq 1$.

Proof. We follow the passage from linear forms to an exponent bound used by Zudilin at integer bases [4, p. 162]. Applying the estimates separately at each fixed base $(a/b)^r$ will give a bound independent of r . We use the polynomials U_n, V_n from Theorem 4.1, with $W_n = \deg U_n$. In that construction the coefficient of F before cancellation is a sum of $O(n)$ Laurent monomials times two Gaussian binomial polynomials. Each Gaussian polynomial has nonnegative coefficients summing to at most 2^{27n+2} . Thus the sum of the absolute coefficients is $\exp(O(n))$. Its largest exponent is $K_n = (1091n^2 + 81n + 2)/2$, so its absolute value at a fixed $x > 1$ is at most $x^{K_n} \exp(O(n))$. The cyclotomic estimate in the same proof bounds the normalising multiplier by $x^{W_n - K_n} \exp(O_x(n))$. Multiplication gives

$$|U_n(x)| \leq x^{W_n} \exp(O_x(n)) \quad (x > 1 \text{ fixed}).$$

Set $\xi = F(a/b)$, $Q_n = b^{W_n} U_n(a/b)$ and $P_n = b^{W_n} V_n(a/b)$. With

$$\alpha = (C_1 - C_0) \log a, \quad \tau = C_0 \log a - C_1 \log b > 0,$$

we have $\log(Q_n \xi - P_n) = -\tau n^2 + o(n^2)$ and $|Q_n| \leq \exp(\alpha n^2 + o(n^2))$.

For integers A, B, p, q with $q > 0$, if $L = A\xi - B$ and $2q|L| \leq 1$, then

$$|L| \leq |A| |\xi - p/q|.$$

When $Ap - Bq = 0$ this is equality. Otherwise the nonzero integer $Ap - Bq$ gives $1/q \leq |L| + |A| |\xi - p/q|$, which proves the inequality. We fix $0 < \eta < \tau$ and use the preceding estimates in the form

$$e^{-(\tau+\eta)n^2} \leq Q_n \xi - P_n \leq e^{-(\tau-\eta)n^2}, \quad |Q_n| \leq e^{(\alpha+\eta)n^2}$$

for all sufficiently large n . Given a sufficiently large denominator q , we choose $n = \lceil \sqrt{\log(2q)/(\tau - \eta)} \rceil$, so that $2q(Q_n \xi - P_n) \leq 1$. The preceding integer argument now applies to $(A, B) = (Q_n, P_n)$ for every numerator p . Also $Q_n \neq 0$: otherwise $Q_n \xi - P_n$ would be an integer strictly between 0 and 1. It follows that

$$|\xi - p/q| \geq e^{-(\alpha + \tau + 2\eta)n^2} = q^{-(\alpha + \tau + 2\eta)/(\tau - \eta) - o(1)},$$

since $n^2 = \log(2q)/(\tau - \eta) + O_\eta(\sqrt{\log q} + 1)$.

Letting $\eta \downarrow 0$, we deduce the bound $1 + \alpha/\tau = (1 - \theta)/(\theta^* - \theta)$. A common power multiplies α and τ by r , so the quotient is unchanged, although the constants in the approximation inequality may depend on r . For $31/4$, the exact interval bounds in the companion record, [Section 2.5](#), give $\theta < 0.4036982$ and $\theta^* > 0.40568$. The quotient increases with θ and decreases with θ^* in this range, so

$$\frac{1 - \theta}{\theta^* - \theta} < \frac{1 - 0.4036982}{0.40568 - 0.4036982} = \frac{2981509}{9909} < 301.$$

The same section gives the rational certificates for these bounds and sharper enclosures of the constants. $\square$

4.1 Comparison with earlier criteria

Bundschuh and Väänänen's Theorem 2 at $\alpha = -1$ [[2](#), p. 177] gives irrationality for $\log b / \log a < \theta_{\text{BV}} := 1/2 - 1/\pi^2$. Their q is the base $a/b > 1$, not its reciprocal. Their logarithmic derivative is $L_q(z) = \sum_{j \geq 1} (q^j + z)^{-1}$, so the value at $z = \alpha = -1$ is exactly $F(a/b)$. The rational height is $h(q) = a$, and their parameter $\lambda = \log h(q) / \log q$ is $1/(1 - \log b / \log a)$, which gives the displayed cutoff. Since $\pi^2 < 10$, one has $\theta_{\text{BV}} < 2/5 < \log 4 / \log 31$, so $31/4$ lies outside that sufficient region. The two sufficient regions differ on $[\theta_{\text{BV}}, \theta^*)$. Zudilin also notes an extension to noninteger rational bases $p = r/s$ for the generalized q -logarithm when $\log |r| > c \log |s|$, with $c > 0$ computable but unspecified [[5](#), Sec. 2, p. 4]. For F , the specialisation above gives $c = \mu$, the exponent bound of [[4](#), p. 162].

The positive-remainder estimates require $x > 1$, so they give no statement for negative bases. The series diverges at $x = 1$. The companion record also checks the parameters at $7/2$ in its section on Bundschuh and Väänänen's height criterion. The analytic irrationality result used there is their cited theorem.

5 Further questions

To use the determinant asymptotic in an irrationality proof, we would need to clear its rational coefficients at a cost compatible with its decay. Positivity and the growth of the moment weights do not estimate this cost. For comparison, the [rational-value model in the companion paper](#) has the same formal orders, leading coefficients and type of fixed-base power correction, although its target value is rational. The model and its denominator calculation are retained there with ordinary proofs as unformalised remarks. The long record also gives an [ordinary argument for the ratio-limit extension](#)

under $a_{k+1}/a_k \rightarrow \rho \in (0, q^{-1})$, without the polynomial shift bound in Theorem 2.1. That extension remains unformalised.

A different question concerns the polynomial coefficients of the approximants, rather than their remainders. Van Assche's little- q -Legendre construction [12, §3] and the multiple-orthogonality construction of Postelmans and Van Assche [13, §2] make the normalisation important. In the 2016 family, the companion paper proves positivity of the first matrix of a coefficient pencil through rank eight, with real, interlacing roots below $F(p)$ for $p > 1$. The shifted minors and coefficientwise positivity are supported by the stated finite polynomial computations. The [coefficient-moment section](#) gives the definitions and the precise ranges. It leaves all-rank positivity and convergence of the largest root open. The general moment criteria of Wang and Zhu [15] and Sokal and Walrad [17], Berg's factorial-power example [16, Theorem 5.1], and the quadrature construction of Golub and Welsch [18] provide the relevant comparisons. Using a recurrence from another family would require a proof that it holds for these coefficients. The root-of-unity method of Krattenthaler, Rochev, Väänänen and Zudilin [14], for example, concerns a separate construction.

5.1 Congruences and divided forms

At $3/2$, a polynomial Q of degree at most W has integral homogenised value $H_W(Q) = 2^W Q(3/2)$. Congruences at powers of 2 and 3 give finite counting criteria for combinations of such values. The companion paper treats [integer rescaling](#) and [endpoint congruences and signed sums](#). It also treats the rank-one case and proves the quantitative finite-fibre estimate used below. Equal residues give a useful irrationality form only when the divided remainder is also nonzero and small. Smith normal form describes the image of the evaluated lattice [6, Theorems 2.3–2.4], but does not estimate these real remainders.

For example, suppose that M integral rows (A_j, B_j) have remainders $e_j = A_j F(3/2) - B_j$, and that the subset sums take Q residue values modulo a positive integer D . Put $T = \sum_j |e_j|$ and suppose that at most k selectors give the same exact remainder within a residue class. For each positive integer n , the sufficient inequality

$$2^M > Qk \left(\left\lfloor \frac{nT}{D} \right\rfloor + 1 \right) \quad (11)$$

gives two equal-residue selectors whose remainders differ by a nonzero quantity of absolute value less than D/n . Indeed, in each residue class partition the interval of attained remainders into bins of width D/n , beginning at its least value. If every bin contained only one real value, its multiplicity would be at most k , contrary to (11). Subtracting the selectors and dividing by D gives an integral linear form of nonzero absolute value less than $1/n$. The [quantitative lemma and its proof](#) state the finite-set version without assumptions about Lambert series.

The elementary integer-base method provides a different comparison. Vandehey [10] and Duverney and Tachiya [11] use integer-base expansions. At a noninteger rational base the cleared tails satisfy a recurrence with a growing denominator contribution. The [clearing and tail-recurrence calculations](#) in the companion paper rule out one coordinatewise scheme at $3/2$. They make no assertion about other approximation families.

At the function level, Bell and Smertnig [9, Theorem 1.3] exclude Mahler equations for the divisor generating series, while leaving the arithmetic of a single rational argument undecided.

A construction problem. The unrestricted request for primitive polynomial rows and a signed combination whose divided remainder lies strictly between 0 and $1/n$ is equivalent to the irrationality of the target. The companion paper retains the full statement and proves this equivalence in its [section on divided linear forms](#). A more specific question must prescribe the approximation family. For the 2004 family one can instead ask whether a parameter direction improves θ^* . The [all-base degree restriction](#) proves the upper bound $1/2$ for the degree cutoff under its stated uniform-in-base hypotheses. Thus this sufficient criterion does not include $3/2$. When the degrees have a quadratic limit, the stronger conclusion there excludes decay of the undivided forms at $3/2$. Division by additional base-dependent content and other families remain unclassified. Equality in the rational-base cutoff is also unresolved by the estimates of Section 4.

A Proofs and source records

The margin links give the existing Lean proofs and recorded Comparator comparisons, indexed in the [evidence record](#) at the [source revision](#). Theorem 2.1 is the polynomial-bound specialisation of the [geometric-moment proof](#) cited in the long record; its margin mark awaits regeneration. The broader ratio-limit argument and rational-value model there have ordinary proofs only. This revision has had no fresh Lean or Comparator run and no independent review.

Data availability. The cited source revision contains the Lean sources, toolchain and library manifest. The companion paper supplies the longer algebraic arguments and the precise scope of the finite computations.

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